Milestones List

Status	Date	Milestone or Completed Deliverable	What it Demonstrates	How it is Verified	Verified by
Complete	9/2/2026	Professional Biography	Demonstrates each team member's educational and technical background and relevant skills and contributions.	Each biography is submitted to canvas by the deadline and reviewed and then uploaded to team repository.	Course Instructor / TA
Complete	9/9/2026	Team Formation and Project Description	Demonstrates that the team has been established and has selected a defined project that is feasible within the time and resources the team has.	The document is submitted on canvas by the deadline which it identifies all members of the group and an outline of the project that is worked on	Team Members and Course Instructor / TA
Complete	9/18/2026	Advisor Selection and Design Constraints Essay	Demonstrates that the team has obtained technical guidance and identified constraints that are appropriate and will influence the design.	The team reached out to the selected advisor, and they agreed to support the project. The submitted essay explains the design constraints and how they may affect decisions.	Project advisor and Course Instructor / TA
Complete	9/28/2026	User Stories and Use Cases	Demonstrates that the team understands the intended users of the project as well as important features and the outcome the system must support.	The user stories that are submitted follow the correct format, flows, actions, and expected outcomes and is reviewed by team before submission.	Team members, Course Instructor / TA
Complete	10/2/2026	High-Level System Design (Design D0)	Demonstrates that the solution proposed by the team has been organized into	The D0 design includes descriptions and plans for major components and	Team members, Course Instructor / TA, Project advisor

			major components, subsystems, and certain interfaces, and all the information addresses the projects' central problem.	their interactions. Before submitting on time to canvas, the team reviews the document to confirm addressing design objectives.	
Complete	10/8/2026	Detailed System Design, Part 1 (Design D1)	Demonstrates that the high-level design has been developed into a more detailed technical solution for the first set of system components.	The D1 submission to canvas includes the important diagrams, interface descriptions, design decisions, and component level details. The team reviews this before submitting on time to canvas.	Team members, Course Instructor/ TA, Project Advisor
Complete	10/14/2026	Detailed System Design, Part 2 (Design D2) and Peer Review Notes	Demonstrates that the remaining system details have been defined, and the complete design has hit a peer review phase to identify conflicts, risks, and necessary revisions.	The D2 submission completes the detailed system design. The peer review notes identify the reviewers and their comments as well as identified issues so the team can review the other documents as well as this one and make necessary changes.	Team members, Peer Reviewers, Course Instructor/ TA, Project Advisor
Planned	10/28/2026	Task List, Milestones, Timeline, and Effort Matrix	Demonstrates that the team has translated the design details and documents into an itemized realistic implementation plan with measurable outcomes and equitable division of labor.	The plan contains the complete task list, a milestone list, a timeline through implementation, effort matrix, and risk register to check for consistency and workload balance.	Team members, course instructor/ TA
Planned	10/30/2026	Requirements Document and Technical	Demonstrates that the users and stakeholder needs	Document has each requirement having a unique id,	Team members, course

		Specification Document	have been met and have been converted into measurable testable requirements and technical specifications	measurable criterion, and verification method. The team will review document before submitting on time to canvas	instructor/TA, project advisor
Planned	11/4/2026	Test and Validation Plan Draft	Demonstrates that the team has defined how the completed system will meet its technical requirements and satisfy users' needs and see what needs altered at this point.	The draft maps the requirements to test cases and identifies their test procedures. Also has explicit pass and fail criteria and verification and validation activities. All members review and are a part of this step to understand the testing.	Team members, course instructor/TA, project advisor
Planned	11/11/2026	Ethics and Responsible-Use Statement	Demonstrates that the team has considered everyone and everything involved and the potential effects of the project on users and stakeholders. These could be privacy, security, and professional responsibility.	The submitted use statement identifies relevant risks to affected stakeholders. This is reviewed against the system requirements, and design decisions and changes are made depending on the comparison and then submitted on time to canvas.	Team members, course instructor/TA, project advisor
Planned	11/18/2026	Presentation Draft and Documented Advisor Feedback	Demonstrates that the whole team has developed an explanation of the project utilizing the advisor's feedback to improve accuracy and clarity of the presentation.	A complete presentation draft is reviewed by the advisor, and the feedback is written and documented and addressed appropriately to allow for the best outcome at the end of the full presentation.	Team members, course instructor/TA, project advisor

Planned	11/18/2026	Team Oral Presentation, Recorded Video, and Peer Reviews	Demonstrates that the team can communicate professionally to address the problem, design, implementation plan, and expected use and value that comes from the project.	The entire team will give a part of the presentation, and the recording is submitted on time on canvas. We will meet the time requirement, and the topics covered requirement, and we will utilize the peer review as well.	Team members, peer reviewers, course instructor/TA, project advisor
Planned	12/2/2026	Design Report, Design Notebook, and Team and Self-Assessment	Demonstrates that the team can properly assess themselves within the team as well as the ability to produce a complete and traceable record of the design process and be prepared for implementation.	The final design report will be checked against rubric for completeness and consistency. The notebook contains all the dated work and evidence over the months of the effort contributed to the project. The team and self-assessments will be submitted by all members.	Team members, course instructor/TA, project advisor
Planned	12/8/2026	Project Advisor Assessments	Demonstrates that an independent professional advisor has evaluated the team on many different metrics and given feedback along the process of the project to allow the team to be ready for project implementation.	The project advisor will complete and submit the required assessment form by the deadline. If there are any concerns or feedback that are holding the team back from implementation, they will get added to the corrective action / risk list.	Team members, course instructor/TA, project advisor

Internal Milestones through Implementation Semester:

Status	Date	Internal Milestone	What it Demonstrates	How it is Verified	Verified by
Planned	1/10/2027	Login/Signup backend logic implemented	Demonstrates that the milestone for the development of the login endpoints and data tables	This can be verified through the unit tests	Team members
Planned	1/13/2027	Image uploader AiClient and service work completed and header started.	Demonstrates that the milestone for AiClient and AI service work will be completed and documented with backup plan.	This is verified through documentation and team member reviewing the choices made.	Team members
Planned	1/14/2027	Switcher backend logic implemented	Demonstrates that this will be the milestone for having the backend logic implemented. This will include the endpoints as well	This can be verified through a series of unit tests that should be made along with them	Team members
Planned	1/22/2027	3D Model Viewer renders a stored model	Demonstrates that a model saved in the system can be opened from its glbUrl and shown in the browser	A model opens from the Switcher's glbUrl, switching between models 10 times shows only the latest one, and browser memory does not keep growing	Team members
Planned	1/25/2027	Front-end login/signup logic	Demonstrates that this will be the milestone for having the front end developed and functional. This won't include styling	This can be verified through user tests	Team members
Planned	1/28/2027	Front-end switcher logic implemented	Demonstrates that this will serve as the milestone for the front end. This will include all functionality need to run without focusing on styling	This will be verified with basic testing, any major defects will be handled then, but small issues will be delayed	Team members

				until complete testing is done.	
Planned	1/31/2027	Upload intake Working	Demonstrates that a user can upload a valid image through the screen and bad uploads and errors are rejected.	Verified by a good upload and a bad upload that gets rejected in the browser, this is also checked by teammates during testing.	Team members
Planned	2/10/2027	Completed testing of Login/Sign-up	Demonstrates that the login/sign-up has been fully tested and passes all edge case tests	Going through the various edge cases created for the section and verifying everything functions as expected	Advisor / Team members
Planned	2/14/2027	Styling of switcher complete	Demonstrates that this will be the point where the switcher will be updated to look as expected by the team.	This will be verified by showing it to our advisor as well as team members coming to appearance over the design.	Advisor/ Team members
Planned	2/17/2027	Generation pipeline working	Demonstrates that an upload becomes a valid GBL with failures handled.	This will be verified by test images to produce valid GLBs. Also testing the GlbValidator and some edge cases like errors happening during the job.	Team members
Planned	2/19/2027	3D Model Viewer feature-complete	Demonstrates that the viewer supports camera controls, layer switching, error messages, and accessibility.	Mouse, touch, and keyboard controls work. Toggling a layer updates without reloading. Each error code shows its message. A screen reader	Team members

				reads the model description (US-05).	
Planned	3/1/2027	Models saved and listed	Demonstrates that a generated model persists and shows up for the user.	This will be verified by a Model row, glb_url, and Layer row exist, and the model appears in the Switcher list.	Team members
Planned	3/12/2027	Image Upload pipeline tested	Demonstrates that every upload edge case has been run, and the must fix failures have been resolved.	This will be verified by a list of test cases that we will run through with the testing phase having a documented pass or fail, and an uploaded PNG will appear in the Switcher and will then open in the Viewer.	Team members
Planned	3/12/2027	3D Model Viewer integrated with the Switcher	Demonstrates that a model selected in the Switcher opens correctly in the Viewer and that every viewer edge case is handled.	Test cases for EC-V-01 to EC-V-09 are run with a documented pass or fail, and must-fix failures are resolved.	Advisor / Team members
Planned	3/20/2027	Completed testing of switcher	Demonstrates that the switcher has been fully tested and verified to be working	This will be verified by a series of tests to ensure the switcher is working and interacting with other components as expected.	Advisor/ Team members
Planned	4/13/2027	CEAS EXPO	Demonstrates the completion of our project	The whole team will be in attendance of the CEAS EXPO	Team members

				to present our finished project	

Timeline - Table 1

Task ID	Statement	Owner	Start Date	Completion Date	Dependencies	Planned Completion Date
T1	Select the AI service and develop AiClient	Abby	1/5/2027		NA	1/13/2027
T2	Develop the upload intake	Abby	1/14/2027		NA	1/31/2027
T3	Develop the GenerationJob worker and GlbValidator	Abby	2/1/2027		T1	2/17/2027
T4	Develop ModelStore.save	Abby	2/18/2027		T3, Switcher list endpoint	3/1/2027
T5	Testing the Image Upload pipeline	Abby	3/2/2027		T2, T3, T4, Switcher, and Viewer working.	3/12/2027
T6	Gather the requirements necessary to start the design and development of the screen/feature	Veronica	1/10/2027		NA	1/12/2027
T7	Design the login/sign-up landing screen	Veronica	1/20/2027		NA	1/21/2027
T8	Develop data table for user information, session data	Veronica	1/11/2027		NA	1/14/2027
T9	Develop login and signup components for the front end	Veronica	1/25/2027		Depends on T8	1/30/2027
T10	Test and validate the edge cases EC-L-01 to EC-L-06	Veronica	2/10/2027		Can't do until T9 is complete	2/25/2027

T11	Baseline code base up and running	Jake	12/15/2026			12/30/2026
T12	Backend for switcher written and functioning	Jake	12/30/2026			1/14/2027
T13	Front end for switcher written and functioning	Jake	1/14/2027			1/28/2027
T14	Styling for switcher is completed	Jake	1/28/2027		Can't do until T13	2/14/2027
T15	Testing for switcher is complete	Jake	2/14/2027		Can't do until T12,T13,T14	3/30/2027
T16	Research and select WebGL rendering library and GLB loader for the 3D Model Viewer	Sophia	11/2/2026		NA	11/13/2026
T17	Develop the 3D Model Viewer's model loading	Sophia	1/11/2027		T16. T4 (GLB file storage; sample GLBs used until ready)	1/22/2027
T18	Develop camera controls and layer visibility for the 3D Model Viewer	Sophia	1/25/2027		T17 (handmade layered test GLB used until T4 saves Layer rows)	2/5/2027
T19	Develop error states and accessibility for the 3D Model Viewer	Sophia	2/8/2027		T17, T18	2/19/2027
T20	Test and validate the 3D Model Viewer's integration with the 3D Model Switcher	Sophia	3/1/2027		T18, T19, T4, Switcher front end (Jake)	2/26/2027
Milestone	CEAS EXPO	Team	NA		All tasks	4/13/2027

Effort Matrix - Table 2

P = Primary Responsibility

Task	Abby	Veronica	Sophia	Jake	Row Total
T1	P - 19	1	1	1	22
T2	P - 24	1	1	1	28
T3	P - 25	1	1	1	28
T4	P - 18	1	1	1	21
T5	P - 16	1	1	1	19
T6	1	P - 8	1	1	11
T7	1	P - 10	1	1	13
T8	1	P - 10	1	1	13
T9	1	P - 8	1	1	11
T10	1	P - 10	1	1	13
T11	1	1	1	P - 10	13
T12	1	1	1	P - 20	23
T13	1	1	1	P - 20	23
T14	1	1	1	P - 10	13
T15	1	1	1	P - 40	43
T16	1	1	P - 9	1	12
T17	1	1	P - 22	1	25
T18	1	1	P - 17	1	20
T19	1	1	P - 17	1	20
T20	1	1	P - 13	1	16
Column Total	116	58	93	100	

Uneven column = Member Veronica. Due to the Login/Sign-up screen being a relatively simple implementation compared to the rest of the tasks, Veronica will assist in tasks that have been found to become more challenging and/or time-consuming. In the event other functionalities need to be broken down more, Veronica will be the first to look into the task assuming nothing is blocking the task.

Work Breakdown Structure

Project: Web-Based Interactive Model Visualizer

Phase 1: Requirements

- Use cases and User stories (User_Stories.md and UC-01)

Phase 2: Design

- Design D0 – High-level System Design
- Design D1 – Detailed System Design, Part 1
- Design D2 – Detailed System Design, Part 2, and Peer Review

Phase 3: Implementation

- T1. Select the AI service and develop AiClient (Abby Kraft)
    - Compare candidate services on cost, rate limits, and license.
    - Run test image and confirm the GLB opens in a glTF viewer.
    - Document the comparison and the backup plan and correct model line in D1 on chosen outcome.
    - Develop AiClient with the 60 second timeout.
    - Add the retry waits of 2, 4, and 8 seconds.
- T2. Develop the upload intake (Abby Kraft)
    - Build the front end React and Ant Design upload screen.
    - Add browser-side format and size checks.
    - Add the disabled button when a job is already in action, add the progress bar, and error popups.
    - Develop the UploadController endpoint with header-byte file checks.
    - Add the session token and the UserID check and the 400, 401, 403, 413, 415 responses.
    - Add downscaling over 4096 px.
- T3. Develop the GenerationJob worker and GlbValidator (Abby Kraft)
    - Creation of job records and status updates.
    - Add timeout and one retry.
    - Keep the job running after a tab closes and retain the uploaded image.
    - Develop header and JSON check and integrate the reused glTF validator.
    - Check and add the 50 MB and 2,000,000 triangle limits and return mesh.
- T4. Develop ModelStore.save (Abby Kraft)
    - Develop and write the GLB to file storage.
    - Create the Model row with glb_url and Layer rows.
    - Add Upload.image_hash duplicate detection.
    - Add three save retries then mark the job as failed and added to logs.
- T6. Gather requirements for Login/Sign-up screen (Veronica Malusky)
    - Compile all the planned requirements from D0 and D2 design diagrams
    - Create a single document for the task requirements for tracking
- T7. Develop login/sign up backend (Veronica Malusky)
    - Develop user tables to store user information and session data
    - Implement error handling for the D2 errors (invalid credentials
    - Create Endpoint for authentication
        - Single endpoint with a bit passed in http request to check if it is a login or signup
- T8. Design Login/Sign-up screen to be implemented (Veronica Malusky)
    - Design overall concept of the design
- T9. Develop Login/Sign-up screen front end (Veronica Malusky)
    - Create Login section – IsSignup = 0 (bit to pass to the shared endpoint)
    - Create Signup section – IsSignup = 1 (bit to pass to the shared endpoint)
    - Implement Front end error handling (empty user/pass)
- T11: Code base framework

    - Develop a basic code base which allows for a platform for the rest of the project
    - Essentially a robust “hello world” page
    - Includes back end and front-end setup
- T12 Switcher backend
    - Develop the backend functionality for the switcher
    - Develop unit tests for the switcher.
- T13: Switcher frontend
    - Develop the frontend functionality for the switcher
    - This will not include making it look pretty but instead focusing on functionality
- T14: Stylize switcher
    - Stylize the switcher to match the expected style of the site
- T16. Research and select the WebGL rendering library and GLB loader (Sophia Munoz)
    - Compare candidate libraries on license, GLB support, performance, and documentation
    - Build a test page that renders Khronos glTF sample models
    - Measure frame rate in Chrome, Firefox, and Safari and record the choice
- T17. Develop the 3D Model Viewer’s model loading (Sophia Munoz)
    - Download the GLB from glbUrl with a 15 second timeout
    - Parse the model, fit the camera to its bounding box, and apply default layers
    - Cancel stale loads when a newer model is selected
    - Free the previous model’s memory before loading the next
- T18. Develop camera controls and layer visibility (Sophia Munoz)
    - Add rotate, zoom, pan, and reset for mouse and touch
    - Add keyboard camera controls
    - Develop setLayerVisibility without reloading the file
- T19. Develop error states and accessibility (Sophia Munoz)
    - Show messages for UNSUPPORTED_BROWSER, LOAD_TIMEOUT, LOAD_FAILED, PARSE_FAILED, MODEL_TOO_LARGE, and CONTEXT_LOST
    - Add Retry and Back to switcher actions
    - Develop getDescription for screen readers (US-05)

Phase 4: Testing

- T5. Testing the Image Upload pipeline (Abby Kraft)
    - Write test cases for each edge case:
        - EC-U-01
        - EC-U-02
        - EC-U-03
        - EC-U-04
        - EC-U-05
        - EC-U-06
        - EC-U-07
        - EC-U-08
        - EC-U-09
    - Run the tests and document pass or fail.

    - The must-fix failures will be fixed.
    - Confirm that an uploaded image appears in the Switcher and opens in the Viewer.
- T10. Testing the Login/Sign-up screen functionality (Veronica Malusky)
    - Write test cases for each edge case:
        - EC-L-01
        - EC-L-02
        - EC-L-03
        - EC-L-04
        - EC-L-05
        - EC-L-06
    - Run each test case and document pass or fail
    - All defects will be documented and fixed accordingly
    - Confirm that the user can sign up and login and their user information pulls to the switcher as intended
- T15. Testing Switcher (Jake Martin)
    - Write test cases for each edge case:
        - EC-S-01
        - EC-S-02
        - EC-S-03
        - EC-S-04
        - EC-S-05
        - EC-S-06
    - Run each test case and document pass or fail
    - All defects will be documented and fixed accordingly
- T20. Test and validate 3D Model Viewer integration with the Switcher (Sophia Munoz)
    - Write test cases for each edge case:
        - EC-V-01
        - EC-V-02
        - EC-V-03
        - EC-V-04
        - EC-V-05
        - EC-V-06
        - EC-V-07
        - EC-V-08
        - EC-V-09
    - Run each test case and document pass or fail
    - All defects will be documented and fixed accordingly

Risk Register

Risk	Category	Likelihood	Impact	Strategy	Mitigation	Owner	Trigger

There is no image-to-3D service returning a usable GLB or output is poor	Technical	Medium	High	Reduce	First T1 and test at least three services. Document backup plan and isolate AiClient so provider can be swapped if necessary.	Abby	No test image produced GLB that opens in glTF viewer by 1/13.
AI-generated GLBs are too large or detailed to render smoothly on laptops and phones	Technical	Medium	High	Reduce	Test the largest sample model in T16. Then enforce the MODEL_TOO_LARGE limit (50 MB / 2,000,000 triangles)	Sophia	A sample model renders below 30 fps or takes over 15s to load in T16 or T20
Layered models aren't available in time to test layer switching, since how layers are made is still open (peer review PR-01)	Technical	High	Medium	Reduce	Build T18 against a handmade test GLB with named parts so it doesn't wait on T4	Sophia	No GLB with Layer rows from T4 by 1/25/2027 (start of T18)
Unauthorized access on system using AI	Technical	Low	High - security	Reduce	Apply access control using Session Tokens with the session data stored in the table	Veronica	No protected login screen with session Tokens and hashing for secure passwords
3d model data leaking for secure data such as brain models	Technical	low	low	Reduce	Apply proper access control so that 3d models cannot be accessed without proper verification, 3d models	Jake	NA

					encrypted to avoid attacks		